

Affordances in UX Design

30

Highlights

- The concept of affordance.
- Five different types of affordances and how each kind is used in UX design.
- Cognitive affordance:
 - How shared conventions affect the interpretation of cognitive affordances.
 - False cognitive affordances.
- Physical affordance.
- Sensory affordance.
- Functional affordance.
- Emotional affordance.
- Putting affordances together in UX design:
 - A UX design checklist of affordances.
- User-created affordances as a wake-up call to designers.

30.1 INTRODUCTION

30.1.1 Acknowledgement of Source

To begin, we gratefully acknowledge the kind permission of Taylor & Francis, Ltd., to use a paper published in *Behavior & Information Technology* (Hartson, 2003) as the primary source of material for this chapter.

30.1.2 The Concept of Affordance

The relevant part of what the dictionary says about “to afford” is that it means to offer, yield, provide, give, or furnish something. For example, a particular window in a house may afford a fine view of the outdoors; the window helps one see that nice view. In UX design, where we focus on helping the user, *a UX affordance is something in a UX design that helps the user do or feel something.*

For a brief discussion of the background and history of affordances, see [Section 33.2](#).

30.1.3 The Importance of Affordance Issues in UX Design

Few topics in HCI or UX have been as misunderstood and misused as the notion of “affordance.” Yet few concepts are as central to effective UX design. In fact, the concept of affordance is one of the most fundamental notions pervading all of UX design. Because affordances are all about helping users do things, you can imagine that the majority of usability and UX problems involve design issues about affordances in designs that don’t help the user as much as they should. Likewise, the majority of HCI and UX design *guidelines* are also about affordances.

30.1.4 Demystifying Affordances

Although a crucially important and powerful concept, the notion of “affordance,” as pointed out by Norman (1999), has suffered misunderstanding and misuse (or perhaps uninformed use) by researchers and practitioners alike and in the literature. Because of this confusion in the field about affordances, the concept can seem mysterious and elusive, so our goal for these chapters is to dispel this mystery and reveal it as an intriguing topic that can be fascinating, fun, and very useful in UX design. For more discussion of the confusion over affordances in the literature, see Section 33.3.

30.1.5 Five Different Kinds of Affordance in UX Design

In an effort to clarify the concept of affordance and how it is used in UX design, I (2003) defined four types of affordances, each of which plays a different role in supporting users during interaction, each reflecting user processes and the kinds of actions users make in task performance. Since that paper, we have added a fifth affordance type, resulting in this set of affordance types (as illustrated in Fig. 30-1):

Table 30-1 contains a summary of these affordance types and their roles in UX design.

For simplicity of understanding, in this chapter, we separate the human senses associated with each kind of affordance. But the human brain integrates the information we get from all our senses and uses each to mediate the other (Obrist et al., 2016; Rosenblum, 2013).

As Gaver (1991, p. 81) says, thinking of affordances in terms of design roles “allows us to consider affordances as properties that can be designed and analyzed in their own terms.” Each type of affordance uses different mechanisms, corresponds to different kinds of user actions, and has different requirements for design and different implications in evaluation and problem diagnosis.

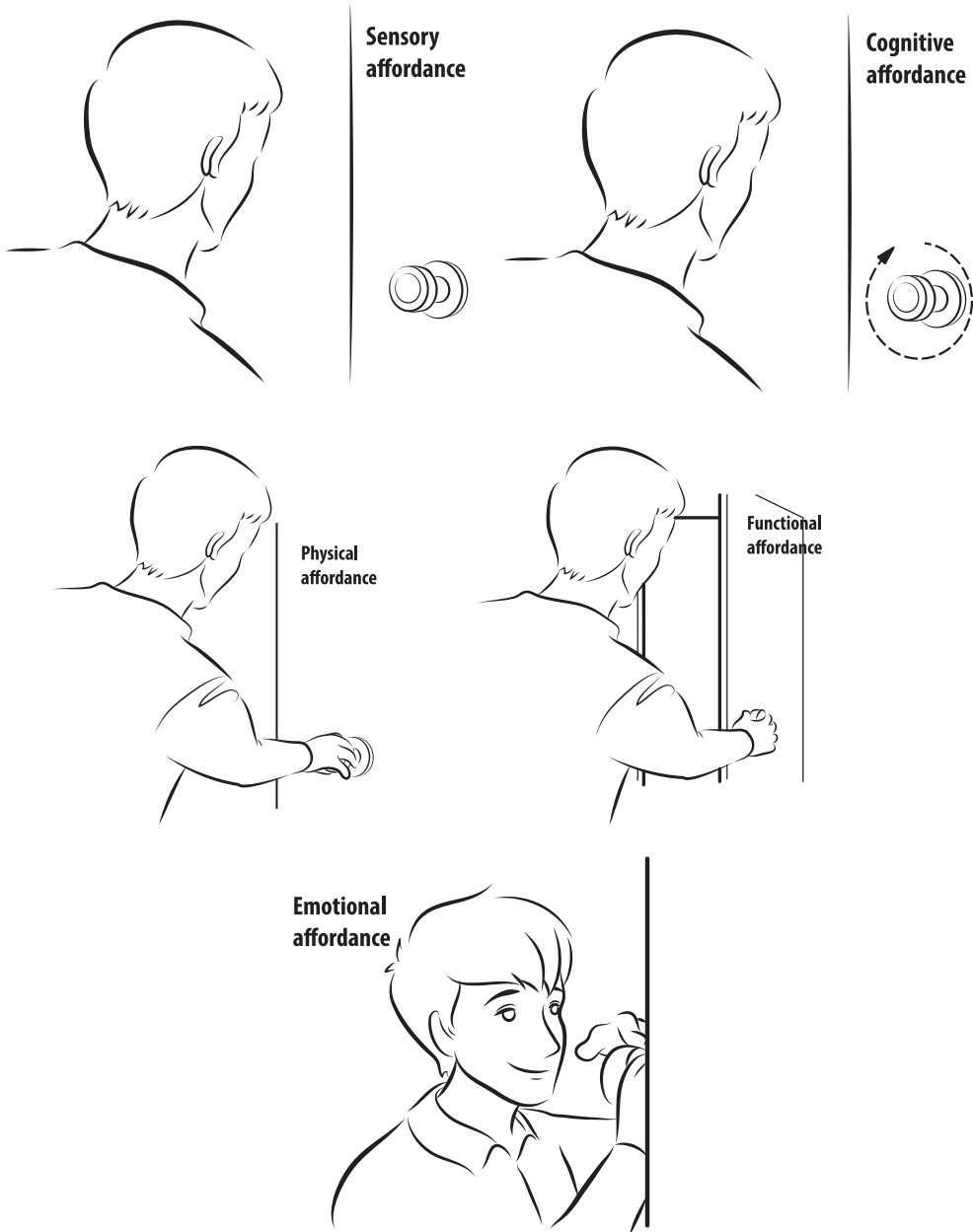


Fig. 30-1
Illustrating the kinds of
affordances in UX design.

Table 30-1
Summary of affordance types

Affordance Type	Description	Example
Cognitive affordance	Design feature that helps users with their cognitive actions: Thinking, deciding, learning, remembering, and knowing about things	A button label that helps users know what will happen if they click on it
Physical affordance	Design feature that helps users with their physical actions: Clicking, touching, pointing, gesturing, and moving things	A button that is large enough so that users can click on it accurately
Sensory affordance	Design feature that helps users with their sensory actions: Seeing, hearing, and feeling (and tasting and smelling) things	A label font size large enough to be discerned
Functional affordance	Design feature that helps users employ a product or system to accomplish work (i.e., usefulness of a system function)	The internal system ability to sort a series of numbers (invoked by users clicking on the Sort button)
Emotional affordance	Design feature that adds emotional impact to the user experience and helps users appreciate and enjoy the interaction	Beautiful aesthetics on a webpage, something that makes interaction fun

As an example to get you started, consider a button available for clicking somewhere on a user interface. Sensory affordance helps you sense (in this case, see or notice) it. Sensory affordance can be supported in design by, for example, the color, size, and/or location of the button. Cognitive affordance helps you understand the button by comprehending what the button is used for, in this case via the meaning of its label. Physical affordance helps you reliably click on this button, so, its design support could include the size of the button or its distance from other related buttons.

30.2 COGNITIVE AFFORDANCES

30.2.1 Introduction

30.2.1.1 Definition of cognitive affordance

A cognitive affordance is a design feature that helps, aids, supports, facilitates, or enables thinking, learning, understanding, and knowing about something. Accordingly, cognitive affordances are among the most significant usage-centered design features in present-day interactive systems, screen based or otherwise. They are the key to answering Norman’s question (1999, p. 39) on behalf of the user: “How do you know what to do?”

Cognitive affordance is associated with the semantics or meaning of user interface artifacts. As a simple example, the symbol of an icon that clearly conveys its meaning could be a cognitive affordance enabling users to understand the icon in terms of the functionality behind it and the consequences of clicking on it. Another cognitive affordance might be in the form of a clear and concise button label.

In this regard, cognitive affordance is used as *feed forward*. It is help with *a priori* knowledge, that is, knowledge used to predict the outcome before making an action on an object such as a button, icon, or menu choice that invokes that functionality.

Another use of cognitive affordance is in *feedback*—helping a user know what happened after a button click, for example, and execution of the corresponding system functionality. Feedback helps users in knowing whether the course of interaction has been successful so far or whether an error has occurred.

30.2.1.2 Starring role in UX design for new users

Cognitive affordances play a starring role in UX design, especially for less experienced users who need help with understanding and learning.

30.2.1.3 How do users acquire cognitive support information?

The whole point of cognitive affordances is that users need knowledge of how things work. What are the ways users can get that needed information about how to use or operate an object or a device?

Knowledge in the head. First, there is knowledge in the head of the user—it gets there by learning through training or experience.

Knowledge in the world. Then there is knowledge in the world—knowledge inherent in objects and devices that gives us clues as to their operation. This is the inherent kind of affordance in Gibson’s ecological view (Gibson, 1977). Gibson was a perceptual psychologist who took an “ecological” approach to perception, meaning he studied the relationship between a living being (let’s say a human being) and its environment, in particular what the environment offers or affords the person. Gibson’s affordances are the properties and objects of the environment as reckoned relative to the person, and “the ‘values’ and ‘meanings’ of things in the environment [that] can be directly perceived” (Gibson, 1977) by the person. Here Gibson is talking about physical properties, giving an example of how a horizontal, flat, and rigid surface affords support for a person for standing or walking.

Knowledge in the design. Finally, there is knowledge that designers add by way of deliberately designed cognitive affordances, such as a label on a button or a message suggesting an action. This communication depends on a shared

understanding between designer and user. See the next section for how we extract meaning from shared cultural conventions.

Example: Mail Icon as Cognitive Affordance

The icon in [Fig. 30-2](#) shows an example of a graphical cognitive affordance. This icon is a clear and succinct way to use a simple image to convey the meaning behind the corresponding key—this is how you get email.



Fig. 30-2
Clear email icon.

30.2.1.4 The meaning of cognitive affordances as found in shared conventions

An important function of cognitive affordance is communication, which depends on agreement about meaning. But often the meaning of a cognitive affordance is not intrinsic to its appearance. The symbols themselves may have no inherent meaning, but a shared convention about the meaning of an image, icon, or symbol (e.g., the shape of a stop sign or an icon for an emergency exit)—even some words or phrases—can convey that meaning.

For examples of how cognitive affordances can be informed by shared cultural conventions, see [Section 33.4](#).

Devices for opening doors

In the tradition of *The Design of Everyday Things* (Norman, 1990), we illustrate with a simple and ubiquitous noncomputer device, a device for opening doors. The hardware store carries both round doorknobs and lever-type door handles. The visual design of both kinds conveys a cognitive affordance, helping users think or know about usage through the implied message their appearance gives to users: “This is what you use to open the door.” The doorknob and lever handle each suggests, in its own way, the grasping and rotating required for operation.

In cases where users must just push on the door to open it, designers often help users understand this by installing, for example, a brass plate to show that one should push and where to push. Even though this plate might also help avoid handprints on the door, it is a cognitive affordance and not a real physical affordance because it adds nothing to the door itself to help the user in the physical part of the pushing action. Sometimes the word “push” is engraved in the plate to augment the clarity of meaning as a cognitive affordance.

But perhaps the message of how to operate a door is so universally understood mainly because of shared conventions. Some people might find nothing intrinsic in the appearance of a doorknob that necessarily conveys this information. On another planet, it could seem mysterious and confusing, but for us, a doorknob is an excellent cognitive affordance because almost all users do share the same easily recognized convention.

Exercise 30-1: Understanding Meaning Based on Cultural Conventions

Can you think of any situation where the meaning of an object, according to shared conventions, is taken for granted in one culture but carries a completely different meaning in another culture? How would these differences affect the behavior of people from the other culture?

30.2.2 Cognitive Affordance Design Issues

Of all affordances, cognitive affordances are probably the most important in UX design and may require the most careful design. We look to cognitive affordances to ensure:

- Clarity of wording.
- Correctness of wording.
- Completeness of wording.
- Predictability of user actions.
- Distinguishability between meanings.
- User centeredness in expressions of meaning.

These cognitive affordance design issues and more will be covered in the UX design guidelines of [Chapter 33](#).

30.2.2.1 Cognitive affordance to get the user started

In [Fig. 30-3](#), we show a nice example of a very simple cognitive affordance that helps early users of Keynote on the Mac know what to do to get started and make a slide for presentation. This application, like its Windows counterpart,

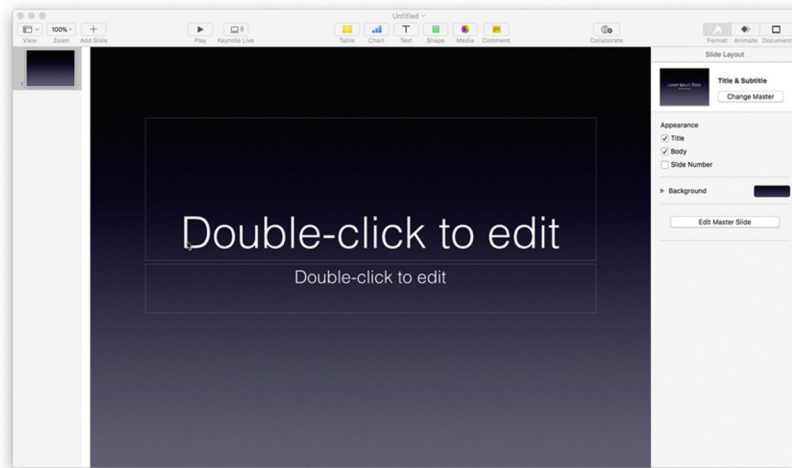


Fig. 30-3

A cognitive affordance to help user get started.

PowerPoint, is typical of large applications where users often start out being faced with a blank screen. All the power of the application is sitting there, but the user doesn't know how to get started. Once you are started, it gets easier because you can respond to feedback in the task thread.

30.2.2.2 Cognitive affordance to help users avoid task completion errors

Sometimes cognitive affordances can help users avoid task completion errors. They're usually in the form of a simple reminder to do the last step, such as "Please take your tickets and your receipt." In Fig. 30-4, we show an example of a cognitive affordance as a reminder to include an email attachment.

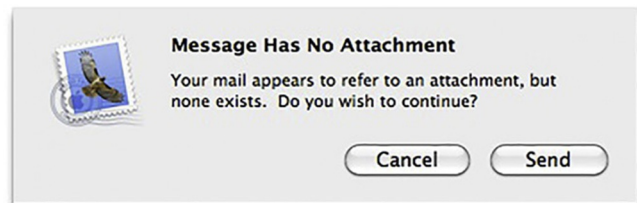


Fig. 30-4

Cognitive affordances used to remind the user to make an email attachment.

If any variation of the word “attach” appears in an email, but it is sent without an attachment, the system asks if the sender intended to attach something.

30.2.2.3 False cognitive affordances misinform and mislead

There are also things that *shouldn't* be in a UX design—things we call *false cognitive affordances*—things that look like cognitive affordances but are wrong and misinform, mislead, and confuse users. Because of the power of cognitive affordances to influence users, misuse of cognitive affordances in design can be a force against usability and a good user experience. Gibson calls this “misinformation in affordances.” As an example, consider a glass panel that appears to be a door but doesn't afford passage. [Draper and Barton \(1993\)](#) call this an “affordance bug.”

Example: False Cognitive Affordance in a Door Sign

What do you think about the picture of a sign on the door of a local store in [Fig. 30-5](#)?

The explanation was that you should use this door only to enter from the outside and not to exit through the door from the inside. They were probably reusing an available design object, the “Do Not Enter” sign, instead of tailoring a sign more specific to the usage situation. The resulting mashup was nonsense.



Fig. 30-5

A door with a confusing sign containing conflicting cognitive affordances.

Example: False Cognitive Affordance in Links Masquerading as Buttons

A common example of a false cognitive affordance is seen in webpage links that are made to look like buttons, but don't behave like buttons. Here's one we found in an evaluation of a real digital library system. The gray background of the text in the top menu bar of a digital library website, [Fig. 30-6](#), makes them seem like buttons acting as tabs from which to pick high-level functionality.



Fig. 30-6

False cognitive affordances in a menu bar with links that look like buttons.

But, in fact, underneath, they are actually just textual hyperlinks, and the gray box around them is just graphical. A user might click on the background, assuming it is part of a button (something we saw more than once in our evaluations), and not get any result. When nothing happened, they were temporarily confused. Because the “button” is actually just a hyperlink, it requires clicking exactly on the text. This is a false cognitive affordance in that it looks like something it is not.

Example: False Cognitive Affordance in a Line That Looks Like the End of a Webpage

Below-the-fold issues on webpages can be compounded by having a horizontal line that happens to fall at the bottom of a screen. Users see the line and assume falsely that it is the bottom of the page and so don't scroll, missing possibly vital information below.

Example: False Cognitive Affordance in a Microwave Control

As another example, [Fig. 30-7](#) is a photo of part of an old microwave.

The dial marks for power settings between the settings for DEFROST and COOK seem to indicate a range of possible settings but, in fact, it is a binary choice: either DEFROST or COOK, with nothing in between. Lots of microwaves



Fig. 30-7
Useless dial marks between power settings on a microwave.

do have many different settings for the power level between defrosting and full-on cooking. But it turns out this one does not. The designer couldn't resist the temptation to fill in the space between these choices with misleading "design details" that are false affordances.

30.3 PHYSICAL AFFORDANCES

30.3.1 Introduction

30.3.1.1 Definition of physical affordance

A physical affordance is a design feature that helps, aids, supports, facilitates, or enables doing something physically. Adequate size and an easy-to-access location could be physical affordance features of an interface button design, enabling users to click easily on the button. Physical actions include clicking, grabbing, dragging, touching, swiping, and so on. Physical affordances are associated with "operability" characteristics of UI artifacts, which can include software artifacts in computer interfaces or they can be real hardware buttons, knobs, handles, dials, or levers. They can be elevator or ATM buttons, or they can even be mechanisms for driving a car, like a steering wheel or brake pedal. Operability is often about physical characteristics of such devices and related human factor issues.

We usually treat active interface objects on a screen, for example, as real physical objects, as they can be on the receiving end of physical actions, such as clicking or dragging.

Example: Add to Cart Button as a Physical Affordance

In Fig. 30-8, we see an example of a physical affordance, the Add to Cart button.

When we analyze the button as a physical affordance, we ask, “How easy is it to click on this button?” The answer, of course, is it is *very* easy to click on this button—after all, they want you to add as much as possible to your cart.



Fig. 30-8

Add to Cart button as physical affordance.

30.3.1.2 Starring role in UX design for experienced or power users

As we said earlier, cognitive affordances play a starring role for novice users. Physical affordances also play a starring role, only for experienced or power users. Their productivity and task performance are all about the speed of physical actions, being able to make physical actions efficient—as rapid and as error-free as possible. Because power users already understand what actions to make, they have less need for cognitive affordances.

30.3.1.3 Some physical affordances are better than others; some depend on the user

The way you decide if one physical affordance is better than another can depend on the individual user. In Fig. 30-9 we show the old BlackBerry phone, one of the few smartphones that had a full hardware keyboard. Some people just liked having a real keyboard like this, one that gives tactile feedback. That is one reason why the makers of BlackBerry have kept it in their design for a long time.

But, of course, on a small device, the keys have to be tiny and close together. Speaking just for me, I think this crowded design makes for less-than-desirable physical affordance and can lead to lots of typing errors.



Fig. 30-9

The BlackBerry mobile phone with a physical keypad.

In contrast, other mobile devices such as the iPhone or iPod have soft touch keypads (Fig. 30-10). I didn't like that as much at first, but it didn't take me long to get used to it, and now, I kind of prefer it. I think the design earned my acceptance through its very good visual and audible feedback. I can see which key I am touching, and when contact is made, and that helps me avoid errors.

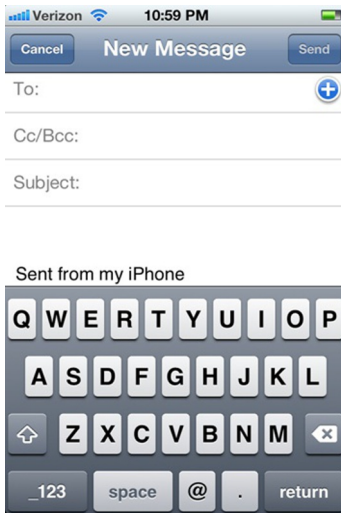


Fig. 30-10

Soft keypad on an old iPhone.

30.3.1.4 Physical affordances for opening doors

Here we return to devices for opening doors. As we look at these in [Fig. 30-11](#), we see that each one is different from the others—different in appearance and even different in how they operate. Each one has its own way to afford physical grasping and rotating for door operation.

Some designs, such as a lever, are considered to give better physical affordance than that of a round knob because the lever is easier to use with slippery hands or with an elbow when the hands are full. The push bar on double doors is another example of a physical affordance helpful to door users with full hands.

Sometimes, the physical affordance to help a user open a door is provided by the door itself; people can open some swinging doors by just pushing on the door. Similarly, sometimes the user of a swinging door must open it by pulling. The door itself doesn't usually offer sufficient physical affordance for the pulling action, so a pull handle is added. A pull handle offers both cognitive and physical affordance, providing a physical means for pulling as well as a visual indication that pulling is required ([Section 30.3.1.5](#)).

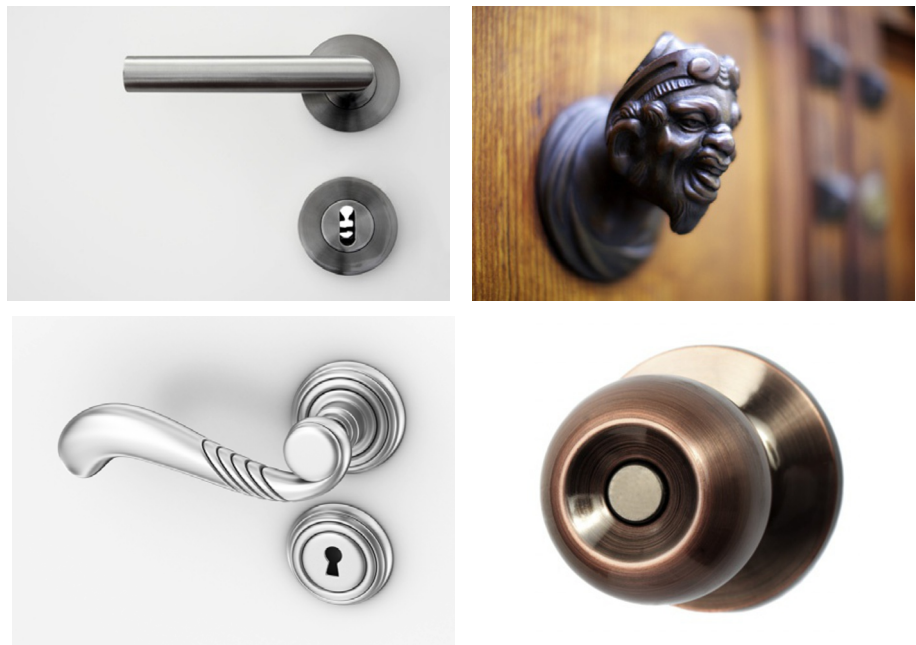


Fig. 30-11
Physical affordances for opening doors.

Example: Some Physical Affordances are Better Than Others: Levers for opening Doors

A number of years ago, I had a chance to design my own home. One of the details I attended to was to be sure that every single doorknob in the house was the “lever” kind that you see in the upper left of [Fig. 30-11](#). That’s because, in my experience, that kind offers a better physical affordance under a wider range of usage conditions. For example, if your hands are full or wet and slippery, the lever-type handle affords operation, say, with your elbow. And people who have arthritis or other physical limitations preventing a solid grasp on a round doorknob, can find it difficult to grab and rotate a knob. The lever works much more easily.

30.3.1.5 Physical devices can also offer cognitive affordance

This kind of physical affordance can also act as a cognitive affordance. As you look at it, its appearance might suggest something about how to operate it. It has its own cognitive affordance built into its physical presence. This is an ecological cognitive affordance, in the J. J. Gibson sense, because the lever and the knob in the [Fig. 30-12](#) in their natural state (without a label, for example) both call out and say, “Grab me, twist me; this is how you open the door.”

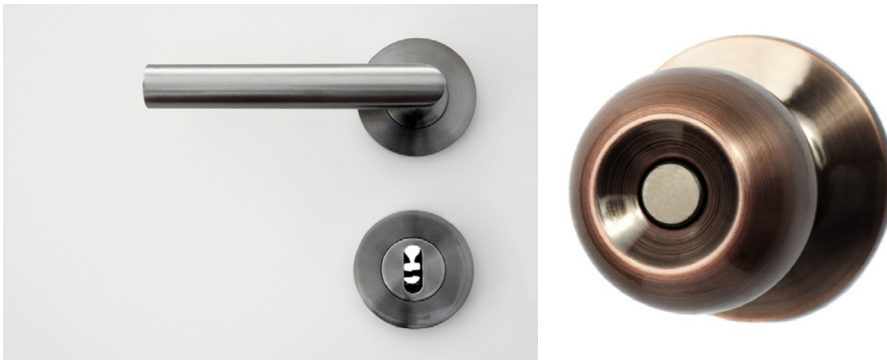


Fig. 30-12
Physical devices as cognitive affordances for how you open a door.

30.3.1.6 Physical devices can also offer emotional affordance

And some designs really *look* good, sometimes so much so that the appearance is a source of aesthetic pleasure and pride. Personally, I like the one in [Fig. 30-13](#). To me, it’s a simple design with beautiful sweeping lines.

It’s up to you whether you think the one in [Fig. 30-14](#) is aesthetic (or grotesque). At least I think it has emotional impact, one way or the other.

Gibson’s Ecological View of Affordances

Perspective on affordances that says knowledge inherent in objects and devices can give us clues as to their operation. Gibson (1977) studied the relationship between a living being (e.g., a human) and its environment, in particular what the environment offers or affords being—for example, how a horizontal rigid surface affords support for a person for standing ([Section 30.2.1.3](#)).



Fig. 30-13
An aesthetic door lever
(according to the author).



Fig. 30-14
Does this door knob elicit
feelings?

30.3.2 Physical Affordance Design Issues

Physical affordance design issues have to do with the following, each of which is explained in subsequent sections:

- Helping users perform physical actions.
- Accommodating physical disabilities.
- Awkwardness of physical actions.
- Human factors and ergonomic issues of device design.

- Physicality.
- Manual dexterity and Fitts' law (Fitts, 1954; MacKenzie, 1992).
- A special characteristic of physical actions that we call physical overshoot.

These physical affordance design issues and more are covered in the UX design guidelines of [Chapter 33](#).

30.3.2.1 *Helping user manipulate objects, do actions*

Above all else, physical affordance design issues are about helping users perform physical actions, usually to manipulate user interface objects. It's about how easy it is for the user to make physical actions. This definitely applies in GUI design, but can also apply to hardware interfaces with real buttons and knobs.

30.3.2.2 *Physical disabilities*

Some human factors and UX designs require attention to physical disabilities and physical limitations of users, and this is at the core of what physical affordances are all about. Clearly, there are individual differences among human users in their physical abilities. For example, very young or older users might have difficulty with fine motor control in using the mouse or another pointing device.

Sometimes, users just naturally have limitations, while others develop disabilities from accidents or disease; it doesn't matter. The design of physical affordances is where you accommodate their needs.

30.3.2.3 *Physical awkwardness*

Awkwardness is something we may not often think about in UX design but, if we do, it can be one of the easiest difficulties to avoid. Designs that present physical awkwardness, such as in holding down the Ctrl, Shift, and Alt keys while dragging with the mouse button down, present an awkwardness that costs users time and energy.

Beyond that, a device that requires an awkward hand movement can lead to *fatigue* in repetitive usage. For example, a touchscreen mounted on the wall at eye level can cause arm fatigue from having to do interactions with one's hand up in the air.

Another example of awkwardness results from the user having to alternate constantly among multiple input devices, such as having to move between keyboard and mouse or between either device and the touchscreen. This kind of behavior involves constant "homing" actions, never getting a chance to "settle down" physically on one device, a time-consuming and effortful distraction of cognitive focus, visual attention, and physical action.

30.3.2.4 Physicality

Physicality is a term referring to real direct physical interaction with real physical (hardware) devices like in the grasping and moving of knobs and levers. This isn't about clicking on "soft" display controls like images of arrows, buttons, or sliders (e.g., to tune a radio or adjust the volume), but is about actually pushing, pulling, grasping, and/or turning real hardware devices such as knobs, buttons, and levers.

Example: Physicality in a Car Shifting Knob

Fig. 30-15 shows a gear-shift knob that is an archetypical example of physicality.

That knob just *looks* so graspable. It's just sticking out there and fits your hand so nicely. The leather exterior gives it a nice texture for grasping. Almost everyone likes the kind of physicality that this thing affords. It gives a sure grip and a satisfying feeling of control as you shift confidently through the gears.



Fig. 30-15

A car gear-shift knob that just says physicality.

Physicality has been an issue for human factors engineers for a long time; Don Norman (2007a) brought it to the attention of the HCI/UX community.

Example: Physicality in Controls for a Radio

Fig. 30-16 shows a car radio that illustrates the issue of physicality.

While this radio does have a physical volume control, there is no knob for tuning. Tuning is done by pushing the up and down arrows on the left-hand side. This design lacks the satisfying kind of physicality you get from the almost universally preferred grasping and turning of a knob.



*Fig. 30-16
A car radio with limited
physicality.*

30.3.2.5 Manual dexterity and Fitts' law

Among the characteristics that affect physical affordance, especially in GUIs, are the size and location of the object to be manipulated. A large object is obviously easier to click on than a tiny one. And location of the object can determine how easy it is to get at the object to manipulate it.

These relationships are represented by Fitts' law, an empirically-based theory expressed as a set of mathematical formulae that govern certain physical actions in human behavior. As it applies in HCI/UX, Fitts' law governs physical movement (e.g., of the cursor) for object selection, moving, dragging, and dropping.

It's specifically about movement from an initial position to a target object at a terminal position. The formulas predict the time to make a movement as:

- Proportional to $\log_2$ of distance moved.
- Inversely proportional to $\log_2$ of target cross-section normal to the direction of motion.

Also (not expressed by Fitts' Law), the time to make a movement is also inversely proportional to the depth of the target along the path of movement.

The potential for errors is the same, namely, proportional to $\log_2$ of the distance moved and inversely proportional to $\log_2$ of target cross-section normal to direction of motion. And, similarly, accuracy is proportional to the depth of the target perpendicular to the path of movement.

In [Table 31-2](#), we translate this to what it means for UX design.

Fitts' law is one of the elements of HCI theory and is the subject of numerous empirical studies in the early HCI literature ([Fitts, 1954](#); [MacKenzie, 1992](#)).

Table 31-2
How Fitts' law plays out in UX design

Practical Conclusions	Design Implications
Longer distance movement takes more time (and produces more fatigue)	Group clickable objects related by task flow close together, but not so close that it could cause erroneous selection
Small objects are harder to click on than large ones	Make selectable objects large enough to be clicked on easily
Smaller (shallower) objects are harder to land on as targets of movement	Make target objects large enough for quick and accurate termination of cursor movement

30.3.2.6 Physical overshoot

What is physical overshoot?

Physical action overshoot is what occurs when you move an object, a cursor, a slider, a lever, a switch—and you go too far in making the physical action, beyond where you wanted to be. An example in a computer interface is the setting of a slider bar, when you pull the slider too far.

Example: Downshifting an Automatic Transmission

Fig. 30-17 shows the transmission gear indicator in an old pickup truck I used to have. Here you can see, from right to left at the bottom, the common linear progression of gears from low (number 1) to high (shown as a circle around “D,” meaning drive or overdrive).

For general driving, it’s a pretty good design, but when you are coming down a long hill, you might want to downshift to third gear on the fly, using engine drag to maintain the speed limit without wearing out the brakes.



Fig. 30-17
Gear shifting with an automatic transmission.

However, because the shifting movement is linear, when you pull that lever down from D, it is too easy to overshoot third gear and end up in second. The result of this error becomes immediately obvious from the engine straining at high RPM.

Exercise 30-2: Other Examples of Physical Overshoot

Can you think of any other examples of physical overshoot that you might encounter in your daily life?

Constraining physical actions to prevent overshoot

In noncomputer examples, you can design to control overshoot by building in physical constraints, as in the example just below. In computer applications, you can ameliorate the risk of overshoot with additional controls, such as using arrow keys to make final slide adjustments.

Example: Constraining a Gear Shift Pattern

Here is an example of introducing a “barrier” that blocks physical movement from going too far, in the shifting mechanism of a Toyota Sienna van, shown in Fig. 30-18.



Fig. 30-18

Constrained gear shifting in a Toyota Sienna van, helping prevent physical overshoot.

Let's say that, while in Drive (4-D in the figure), the driver pulls the lever down to downshift. The lever moves easily down to third gear (3 in the figure) but is blocked by the cutout “template” surrounding the shift lever, preventing it from overshooting and going down to second gear. It is very unlikely that a user will accidentally shift into second gear because that requires an extra action of moving the lever to the right and then further down.

So, here we have a case of consciously taking physical affordance into account in a design by using constraints on physical actions that help users stay within physical boundaries.

30.4 SENSORY AFFORDANCE

30.4.1 Introduction

30.4.1.1 Definition of sensory affordance

A sensory affordance is a design feature that helps, aids, supports, facilitates, or enables user in sensing (e.g., seeing, hearing, feeling) something. Sensory affordance is associated with the “sense-ability” characteristics of user interface artifacts—how well human users can sense (e.g., see) a given interaction object, especially a cognitive, physical, or emotional affordance. Design issues for sensory affordances are about the presentation of user interface objects and include noticeability, discernibility, legibility (in the case of text), and audibility (in the case of sound) of features or devices associated with visual, auditory, haptic/tactile, or other sensations.

While cognitive affordance and physical affordance are the stars of UX design, sensory affordance plays an important supporting role. As an example, the legibility of label text on a button is supported by an adequate size font and appropriate color contrast between text and background, so the label can be read.

30.4.2 Visual Sensory Affordance Design Issues

Visual sensory affordance design issues are about characteristics relating to the user's ability to sense visually (i.e., to see), which include:

- Visibility.
- Noticeability.
- Discernibility.
- Text legibility.
- Distinguishability.

Haptics

UX design issues that entail touch and feel in tactile interaction involving physical contact between user and machine (Section 30.4.4).

- Color.
- Presentation timing.

These sensory affordance design issues and more are covered in the UX design guidelines of [Chapter 33](#).

30.4.2.1 Visibility

The characteristic of visibility is about whether an object important to the user is visible to the user. Sometimes when an object isn't visible, it's because it is blocked by another object. Or sometimes you can't see an object because it simply isn't on the screen. In such cases, the user may have to take some actions to access the object and bring it onto the screen.

Example: Store User Cannot Find the Deodorant

The setting for this noncomputer example is a local grocery store, where a shopper has the simple task of buying some deodorant. The store had been reorganized, so he couldn't rely on his memory to locate the deodorant. He went through the aisles, looking at overhead signs but nothing matched his search goal.

Upon his asking the clerk, he was told, "Oh, it is right over there," being pointed to one of the upfront aisles that he had just scanned. "But I don't see any sign for deodorant." "Oh, yeah, there is a sign," the clerk replied, "you just have to get up real close and look right behind that panel on the top of the end shelf." [Fig. 30-19](#) shows what that panel looked like to someone scanning these upfront aisles.

In [Fig. 30-20](#), you can see the "deodorant" sign revealed if you "just get up real close and look right behind that panel on the top of the end shelf."

The panels on top of the shelves are an aesthetic touch, but they were put in a location that exactly blocked the "deodorant" sign and others, rendering that important cognitive affordance invisible from most perspectives in the store.

30.4.2.2 Noticeability

Suppose an object important to the user is technically visible but not *noticeable*? Being somewhere on the screen isn't necessarily enough for an interaction object. For it to be useful, the user has to notice it. Noticeability is especially important in cases where the user might not even know the needed cognitive affordance exists and isn't necessarily looking for it. It's your job, as the UX



Fig. 30-19
Aesthetic panel blocks
visibility of sign as cognitive
affordance.



Fig. 30-20
The sign is visible if you look
carefully.

designer, to make it so the user will notice it. Your design has to foster user awareness of a necessary object.

The user's focus of attention

The most important design factor in getting an interaction object noticed is putting it right within their *focus* of attention.

A good computer example of an object that often suffers from poor noticeability is a status message line, say, at the bottom of the screen. Such message lines are notoriously unnoticeable because they are outside of most users' focus of attention.

At any point in time within task performance, a user typically has a narrow focus of attention, usually near where the cursor is located. So, a pop-up message next to the cursor will usually be far more noticeable than a message in a line at the bottom of the screen.

Object size

Larger objects are easier to notice.

Object color contrast and separation from clutter

Object color that makes good contrast with the background can make an interaction object noticeable. Another important factor affecting noticeability is the separation of the object of interest from the clutter of other user interface objects.

Example: Where is the “Start Presentation” Icon?

PowerPoint users with any experience know that the icon for getting into the presentation mode is the little icon that looks like a slide show screen. And, once you know where it is, it's no problem. But the inexperienced user might never find it without help because this icon is so small, and it's hidden.

In the partial screen image in [Fig. 30-21](#) of a PowerPoint slide being edited, the red arrow points to the very tiny little “Start Presentation” icon—and yet showing the presentation is one of the most important actions in using PowerPoint!

30.4.2.3 Discernibility

It's not enough to make an important interaction object visible and noticeable; it also needs to be discernible. Can the user make out, detect, or recognize an important object? That is, can the user recognize the object, its shapes, and colors?

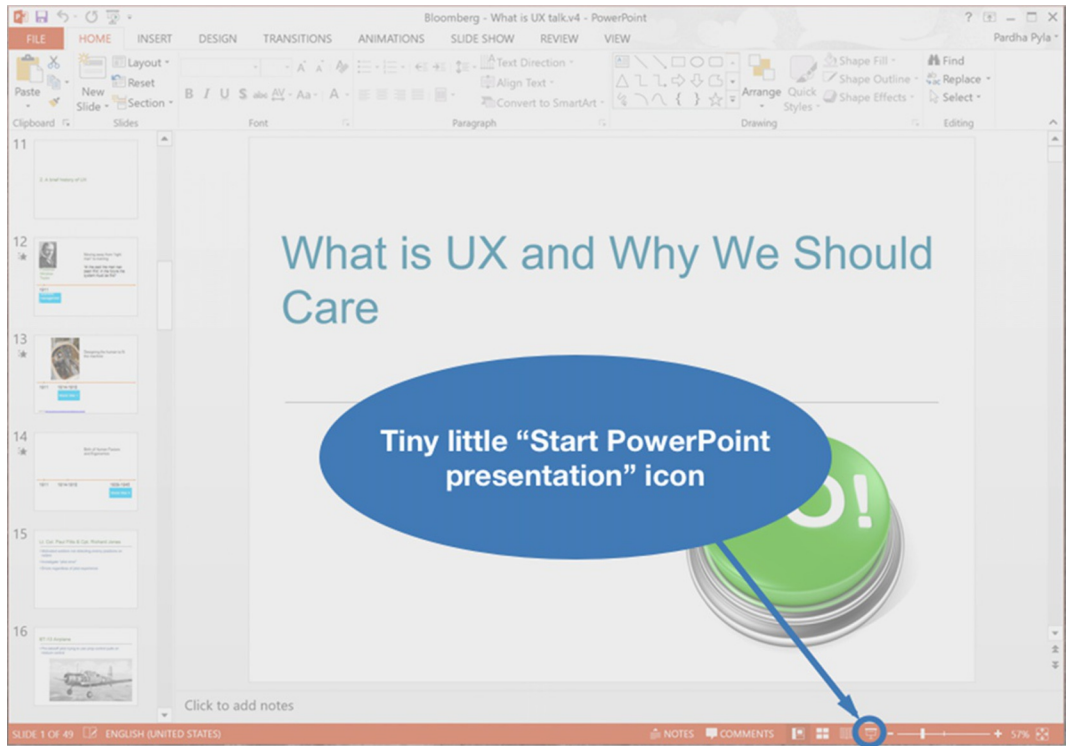


Fig. 30-21

How do you start a PowerPoint presentation?

Example: Black-on-Black in Stereo

Fig. 30-22 shows a picture of a panel on a CD player that used to be part of my stereo system. The controls, such as for play and stop, are black buttons embossed with black icons. Some designer must have thought it would be cool and aesthetic to use this black-on-black color scheme.

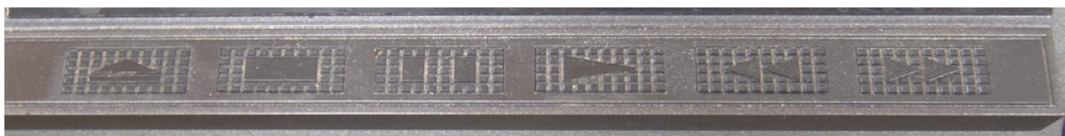


Fig. 30-22

Fashionable (?) design of gray-black icons on a gray-black background.

But these symbols are not easily discernible. You can't make them out very well visually, even in a good light. And the symbols aren't raised enough to be discernible as tactile inputs.

This is also a good example about designing for usage in context. Often you want to lower the lights and kick back while you're listening to music. While this may make for a relaxing ambience, it does nothing to help the visibility and discernibility of the symbols on these controls.

30.4.2.4 Text legibility

Legibility is about discernibility of text. Can the user make out text in order to read it? Legibility is about the text being presented so it can be read or sensed, not about whether the meaning of the words is understandable. The meaning is the same regardless of legibility.

The design factors affecting text legibility are obvious and include font type, size, color, and contrast.

Example: Small Font in Crowded Text, Poor Contrast With Background

Look at the sign in [Fig. 30-23](#). This example shows a design (apparently by the systems people) using yellow text on a light background. And someone (a UX person?) left a clarifying note!

30.4.2.5 Distinguishability

Distinguishability applies to all the senses. It's about whether similar but different objects are recognizable as distinct. It's not enough to make an important interaction object visible, noticeable, and discernible. To avoid errors in operating on the wrong object, the interaction object of interest also needs to be distinguishable from other objects.

Example: Here's Soap in Your Eyes

Here is a good example of visual distinguishability and taking usage context into account. In the context of taking a shower, consider the two bottles in [Fig. 30-24](#), one for shampoo and one for conditioner. These are different bottles for different purposes but they are so strikingly similar that users cannot easily tell them apart.

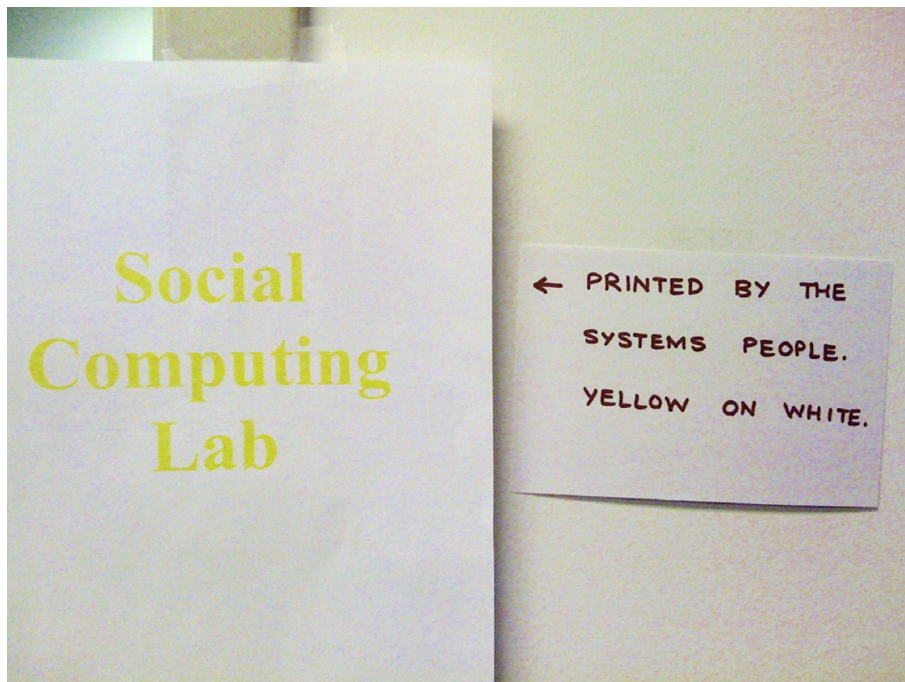


Fig. 30-23

Yellow text on a light background (a real sign on the door of a lab, credit unknown).



Fig. 30-24

Bottles for shampoo and conditioner are difficult to distinguish.

The important distinguishing labels, for shampoo and for conditioner, are “hidden” within other text and are in a very small font—indistinguishable, especially for users with soap in their eyes. And, remember, users who need eyeglasses cannot wear them in the shower.

So, users sometimes add their own affordances. In the case shown in Fig. 30-25, a user has added labels on the tops of these bottles to tell shampoo from conditioner in the shower.



User-Created Affordance

An affordance, usually a cognitive or physical affordance, that is added to the original design (for example, by taping on a label) by a user who needs that affordance for a better user experience and to enhance the design (Section 30.8).

Fig. 30-25

Because of poor sensory affordance, a user had to mark the bottles to tell them apart.

You can see, in Fig. 30-26, an example of a kind of shampoo bottle design that would have avoided the problem in the first place.



Fig. 30-26

Better: A design to distinguish the bottles.

In this clever design, the shampoo, the first one you need, is right-side up, and the labeling on the conditioner bottle, the next one you need, is printed so that you stand the bottle “upside down.”

30.4.2.6 Color

The use of color is an interesting facet of visual sensory design. Color decisions are often made by specialist graphic designers and constrained by organizational standards and branding requirements. For a bit more about guidelines for the use of color in UX design, see [Section 32.10.7](#).

30.4.2.7 Presentation timing

Sometimes a user needs an affordance at a certain time during the interaction. In these cases, it just doesn’t work if it isn’t there when you need it.

Example: Just-In-Time Towel Dispenser Message

[Fig. 30-27](#) is a photograph of a paper towel dispenser in a public bathroom. The next available towel is visible, and the cognitive affordance in the sketch on the cover of the dispenser clearly says “Pull the towel down with both hands.”



Fig. 30-27

The primary cognitive affordance for taking a paper towel.

But what if the next towel fails to drop down and users cannot grab it? Now a different user action is needed to get a towel, so a different cognitive affordance is required, as illustrated in [Fig. 30-28](#).



Fig. 30-28

The backup cognitive affordance to help start a new paper towel.

Designers provided this new cognitive affordance, telling the user to push the lever to get the next towel started down into position. Normally this second cognitive affordance is obscured by the towel, but it becomes visible just when needed.

From the “wear marks” on the push bar, you can see that this “*exception*” case, and the need for this added cognitive affordance, have occurred quite often.

30.4.3 Auditory Sensory Issues

While the majority of objects in UX designs are visual, there are also auditory signals to users. Auditory sensory design issues are somewhat parallel to and similar to the *visual* sensory design issues, only they have to do with sound and how well users are able to hear sounds. Here is a summary of auditory issues for UX design.

- Audibility issues (counterpart to visibility issues):
 - Can users hear the sound?
- Noticeability of audio cues:
 - Volume of auditory cues (a major factor in noticeability).
 - Plus, ability of user to control volume.
- Intelligibility, discernibility, distinguishability of audio:
 - Has much to do with audio quality.
 - Can the user make out what is being said?

Interaction Channel

A means or mode or medium through which users and parts of a system interact and communicate, including sensory modes such as visual, voice, and tactile interaction. The concept also includes devices such as desktop computers and smartphones and system-oriented channels such as the Internet, Wi-Fi connections, and even Bluetooth (Section 16.2.4).

- Frequency, pitch (of sound):
 - Verbal messages should be centered in the mid-range of human hearing.
- Garbling.
- Interference from background noise.
- Timing of audio cue.
- Tone of audio cues.
 - Gentle tone can encourage, reduce stress.
 - Harsh tone can annoy, irritate.
 - But harsh tone can be used to get attention.
- Multiple audio cues can fragment user attention.
- Like visual branding, some audio can be subject to branding constraints.
 - Example: Iconic audio, such as the Microsoft chime at Windows startup.

Interaction designs involving auditory inputs and outputs are becoming more popular. [Patel and Hughes \(2012\)](#) make the case for moving beyond purely visual user interfaces to distribute tasks over other senses. They propose using auditory channels, in particular, to support the task of understanding of meaning and for decreasing cognitive load. One major advantage of auditory interaction components is that sound usually doesn't distract user attention from the visual channel.

They show that nonspeech sound, what they call sonification, is feasible as an interaction channel in certain situations. The challenge in making the sounds meaningful is designing them to be distinguishable in terms of pitch, intensity, and tempo.

30.4.4 Haptic and Tactile Sensory Issues

Issues of haptics and touch have been around in UX design discussions for a long time and, like auditory interactions, they are being used more often. An interesting example of one such design is the touchable virtual object, as manifest in a touchable hologram ([Kugler, 2015](#)). In this approach to haptic interaction, ultrasound is used to focus a shape that can be felt by the human hand.

Haptic and tactile issues (about touch and feel) are somewhat parallel and similar to visual and auditory issues, including:

- Noticeability: Is a tactile signal strong enough for users to feel it?
- Discernibility: Can users make out what a given tactile signal is?
- Distinguishability: Are there enough differences between two possibly similar tactile signals for users to tell the difference?

- Consistency: Are tactile and haptic representations consistent in meaning?
- Tone: As with other senses, we have to design tactile and haptic stimuli not to be harsh or irritating.

The factors of noticeability, discernibility, and distinguishability are often aided through some form of vibration (Thompson & Vandenbroucke, 2015) rather than just tactile pressure. One area where design work using tactile vibration is thriving is automotive user interfaces (Kern & Pfleging, 2013). Haptic feedback is a good match to the physical user actions needed for driving a car. Various kinds of vibrations felt in the steering wheel, the brake pedal, or the driver's seat allow users to receive feedback without having to look away from the driving task. These are especially effective for alerting the driver of situations needing immediate attention, such as an impending accident or drifting from the lane on a road.

30.5 FUNCTIONAL AFFORDANCE

30.5.1 Definition of Functional Affordance

A functional affordance is a design feature that helps users get work done by connecting physical user actions to system, or backend, functionality. Functional affordances are the reason why users will make an action; they want access to the functionality to do something. A functional affordance connects users to the usefulness component of the user experience via the power of the software behind the user interface.

For example, if you look at the button in Fig. 30-29 that says Add to cart, the functional affordance associated with that button is seated in the backend



Fig. 30-29

An Add to Cart button viewed as a physical affordance that leads to a functional affordance in the backend software.

application software that carries out the function implied by that button label—in this case, it puts the item selected for purchase into the virtual shopping cart.

For a discussion about how functional affordances fit in with Gibson’s ecological view, see [Section 33.5](#).

30.6 EMOTIONAL AFFORDANCE

30.6.1 Definition of Emotional Affordance

An emotional affordance is a design feature that helps a user make an emotional connection resulting in emotional impact within the user experience. Emotional affordances include design features that connect to our intuitive appreciation of fun, aesthetics, to our sense of identity, and challenges to personal growth.

Many UX designs meet all the functional requirements but are not very exciting or interesting. They tick all the boxes but don’t light up the lights. The difference is emotional impact, which has become an important component of user experience in most UX designs. The mobile device world is leading attempts to leverage emotional affordances to attract customers. It is especially what new electronic products—like smartphones and personal music players—are all about. And some are proposing computers and devices that can sense and understand the emotions of users ([Thompson, 2010](#)).

Emotional impact is also a central concept in the automobile design world. Manufacturers of cars today compete over the pleasure of driving and the fun of using their electronic entertainment systems.

Example: Emotional Impact in the Hibu Image

In an advertisement for a Website Design service, hibu (see hibu.com) shows nothing but butterflies, saying, “Websites so stunning, they’ll give you butterflies” ([Fig. 30-30](#)).

This is an ad for a high-tech web design company whose motto is “Made for business,” and butterflies are the only image? The message is about design, emotional impact, and beauty! Their text begins with, “Our incredibly beautiful websites...” What a testimony to the importance of emotional impact in their view of UX Design.



Fig. 30-30

A hibu advertisement promising butterflies in the user experience (with permission).

30.6.2 Affordances to Support Meaningfulness

Meaningfulness is a phenomenon through which a product or artifact becomes meaningful in the life of a user. Meaningfulness comes out of emotional impact that develops into a personal relationship with the product, enduring over time. It is difficult to isolate a single design feature that instills meaningfulness; much of that long-term personal feeling of companionship, for example, can be traced to the conceptual design and to the daily usefulness of the product.

Meaningfulness

A personal relationship that develops and endures over time between human users and a product that has become a part of the user's lifestyle (Section 1.4.5).

30.7 PUTTING AFFORDANCES TOGETHER IN DESIGN

User work goals that require affordances to work together (Fig. 30-31):

- Users must *understand* the interaction objects (through cognitive affordances).
- Users must be able to *operate* the interaction objects (through physical affordances).
- Users must be able to *sense* interaction objects (through sensory affordances).
- Users must have *access* to functionality they need (through functional affordances).
- And it would be nice to *engender* a good emotional experience in the process (through emotional affordances).

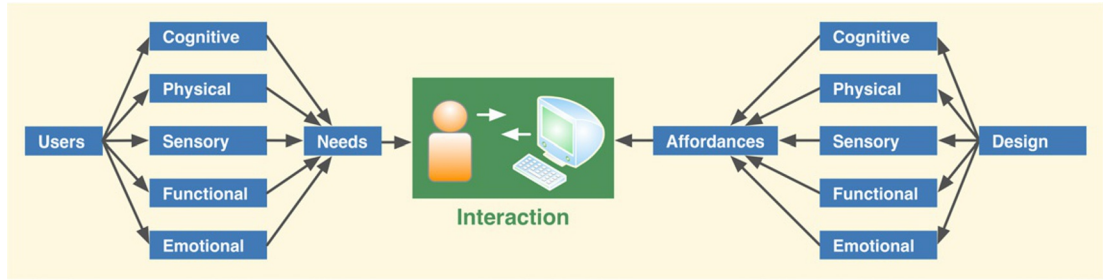


Fig. 30-31

User interaction needs relate directly to affordances in design.

Interaction Artifact

A system, device, service, instrument, mechanism, object, or environment that can communicate with a human, in either or both directions. May include a building or room, a chair, a kitchen, an ATM, an elevator, appliances like refrigerators, cars and other vehicles, most kinds of signage, homes, the workflow of the DMV, and voting machines (Section 1.1).

30.7.1 Affordance Roles—An Alliance in Design

In most UX designs, the five types of affordance work together to help users accomplish work goals by sensing, understanding, and using affordances within a UX design. Each kind of affordance plays a different role in the design of specific attributes of the same artifact, including design of appearance, content, and manipulation characteristics.

An approach to UX design that integrates the different kinds of affordances compels us to consider all five affordance roles together in the design of an interaction artifact.

- Is the functionality to which this interaction object or artifact gives access useful in achieving user goals through task performance (functional affordance, or purpose of physical affordance)?
- Does the design include clear, understandable cues about how to use the artifact (cognitive affordance) or about system outcomes if the artifact is a feedback message?
- Can users easily sense visual (or other) cues about artifact operation (sensory affordance in support of cognitive affordance)?
- Is the artifact easy to manipulate by all users in the target user classes (physical affordance)?
- Can users easily sense the artifact for manipulation (sensory affordance in support of physical affordance)?

Considering one affordance role but ignoring another is likely to result in a flawed design. For example, if the wording for a feedback message is carefully crafted to be clear, complete, and helpful (good cognitive affordance), but users don't notice the message because it is displayed out of the users' focus of attention (poor

sensory affordance), or users cannot read it because the font is too small, the net design is ineffective. A powerful drag-and-drop mechanism on a Mac may offer good physical and functional affordance for opening files (by dragging a file on to an application Icon on the Dock), but lack of a sufficient cognitive affordance to show how it works could mean that most users will not use it.

30.7.2 A UX Design Checklist of Affordances

The set of affordances will serve you well as a kind of UX design checklist (Fig. 30-32), against which each part of your design is checked.

Example: Affordances for a Sort Capability

Let's see how the checklist works with a very simple example, the design of a button to invoke, say, some kind of *Sort* function.

Functional affordance considerations

The first item on the checklist is functional affordance. The designer should ask if the intended functionality, the functional affordance, is appropriate, useful, and available to the user. If not, none of the other affordances will be useful in the end. Let's say functional affordance is OK, so we check this item off our list.

Cognitive affordance considerations

The next item on the list is cognitive affordance considerations. Is the meaning expressed clearly, in this case, in the label or menu choice that invokes the *Sort* function? Does the label advertise the purpose of the button by ensuring its meaning (in terms of a task-oriented view of its underlying functionality) is clear, unambiguous, and complete to help the user know when it is appropriate to click on the button?

As we say in [Section 32.6.3.9](#), long labels are not necessarily bad. Short labels can be nice, but sometimes it takes more words to convey the full meaning. It's also nice to have both a verb and a noun—to indicate the action and the object.

- ✓ Functional affordance
- ✓ Cognitive affordance
- ✓ Physical affordance
- ✓ Sensory affordance
- ✓ Emotional affordance

Fig. 30-32

A UX design affordance checklist.

So, instead of just *Sort*, we could expand it to *Sort Records*. Sometimes even adding an adjective will help convey the meaning more fully, in this case giving us, *Sort Address Records*. Or an adverbial phrase might add information about how the sort will be done: *Sort By Last Name*.

Now, we can check cognitive affordance off the list.

Physical affordance considerations

The designer is next led to consider how physical affordance is to be supported in the button design. Does our *Sort* button design provide adequate physical affordance for easy clicking? Is the size of the button large enough to click on it easily? The button at the left of Fig. 30-33 is large enough so that a little accidental cursor movement during clicking won't move it off the button. But the button on the right is perhaps too small.

Is the button located near other artifacts used in the same and related tasks to minimize mouse movement between task actions? Is the button located far enough away from other, nonrelated, user interface objects to avoid clicking on them erroneously? The button on the left in Fig. 30-33 is isolated from others to reduce errors of clicking on the wrong ones. The button on the right is too close to ones you don't want to click on.

This is also the time to check on physical disabilities. You may not have thought of this, but even some people who don't appear to have physical disabilities may have trouble clicking.

Now, we can check physical affordances off our list.

Sensory affordance considerations

Now, the designer is asked to consider sensory affordance in support of both cognitive affordance and physical affordance in the button design.

In support of the physical affordance, the button must be a color, size, and shape that make it noticeable and, if possible, it should be located in the screen layout so that it is near enough to the user's focus of attention.

In support of cognitive affordance, the button label must have effective font size and color contrast, for example, to help the user discern the label text to read it.

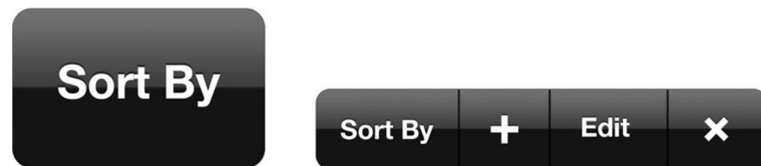


Fig. 30-33

Different buttons have different physical affordances.

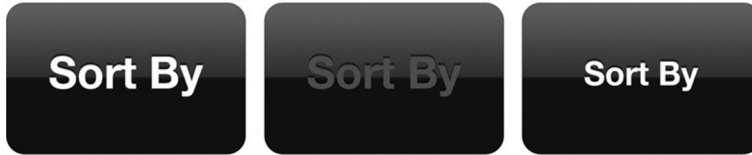


Fig. 30-34

*Different **Sort By** buttons with different sensory affordances.*

On the left side of [Fig. 30-34](#) we have a nice **Sort By** button. But the low-contrast label in the middle is more difficult to read. And the font at the right-hand side is unnecessarily small. There's plenty of space; use it!

Now we can check sensory affordances off our list.

Emotional affordance considerations

Finally, we ask about emotional affordances in this design. Well, it's a little hard to get emotional about a **Sort** button or **Sort** function, I have to admit. But maybe you could make it a little more fun with an animation that shows how the sort works. Novice users might be intrigued by this. Or maybe you could find some interesting way to use sound. Challenge yourself and be creative!

In any case, let's check this off the list.

Exercise 30-3: Affordance Design Checklist

Now that we have shown you how to use a checklist of affordance types in the consideration of a design for a **Sort** button, your job in this exercise is to explain how to apply the same checklist when designing a feedback message indicating a "user error."

30.8 USER-CREATED AFFORDANCES AS A WAKE-UP CALL TO DESIGNERS

A user-created affordance (usually a cognitive affordance or physical affordance) is an affordance that is added to the original design (for example, by taping on a label) by a user who needs that affordance for a better user experience.

If a device in the everyday world doesn't suit the user, we will frequently see users modify the apparatus, briefly and unknowingly switching to the role of designer. We have all seen the little cognitive or physical affordances added to devices by users—Post-it notes added to a computer monitor or keyboard or a better grip taped to the handle of something. These trails of user-created artifacts blazed in the process of day-to-day usage are like wake-up messages, telling designers what the users think they missed in the design.

A common example of trails (literally) of user-made artifacts is seen in the paths worn by people as they walk. Sidewalk designers usually like to make the sidewalk patterns regular, symmetric, and rectilinear. However, the most efficient paths for people getting from one place to the other are often less tidy but more direct. Wear patterns in the grass show where people need or want to walk and, thus, where the sidewalks probably should have been located. The rare and creative sidewalk designer will wait until seeing the worn paths, employing the user-made artifacts as clues to drive the design.

Example: Confusing Glass Door

In [Fig. 30-35](#), a photo of a glass door in a convenience store, we show an example of an added cognitive affordance. The glass and stainless-steel design is elegant: the perfectly symmetric layout and virtually unnoticeable hinges contribute to the uncluttered aesthetic appearance (good emotional affordance), but these same attributes work against cognitive affordance for its operation.

The storeowner noticed many people unsure about which side of the stainless-steel bar to push or pull to open the door, often trying the wrong side first. To help his customers with what should have been an easy task in the first place, he glued a bright yellow cardboard arrow to the glass, pointing out the correct place to operate the door.



Fig. 30-35

Glass door with a user-added cognitive affordance (arrow) indicating proper operation.

Example: Copier Darkness Icon

As another example of user-created affordances, consider the icons shown in [Fig. 30-36](#). These are for lightness and darkness settings on a home office copier, icons with ambiguous meanings. The icon for lighter copies is whiter, but the white can be interpreted as part of the copy, and it seems denser than the icon for darker copies, so the user had to add his own label, as you can see in the figure.

These trails of often inelegant but usually effective artifacts added by frustrated users leave a record of affordance improvements that designers should consider for all their users. Perhaps if designers of the everyday things that [Norman \(1990\)](#) discusses had included usability testing in the field, they would have found these problems before the products went to market.

In the software world, most applications have only very limited capabilities for users to set their preferences. Would not it be much nicer for software users if they could modify UX designs as easily as applying a little duct tape, a Post-it, or extra paint here and there?



Fig. 30-36

A user-created cognitive affordance explaining copier darkness settings.

Example: Car Drink Holder

In [Fig. 30-37](#), we show how a car owner created an artifact to replace an inadequate physical affordance—a built-in drink holder that was too small for today's super-sized drinks. During one trip, the user improvised with a shoe, resulting in this interesting example of a user-installed artifact.

Fig. 30-37

A user-made automobile cup-holder artifact, used with permission from Roundel magazine, BMW Car Club of America, Inc. (Howarth, 2002).



Example: Letterhead Printing

Consider a desktop printer used only occasionally to print on letterhead stationery. Inserting the stationery on top of the existing plain paper supply in the printer does this rather easily. The only problem is that it isn't easy to determine the correct orientation of the sheet to be inserted because:

- There is no clear mental model of how the sheet travels through in the interior mechanism of the printer.
- Printers can vary in this configuration.
- The design of the printer itself gives no cognitive affordance for loading a single sheet of letterhead.

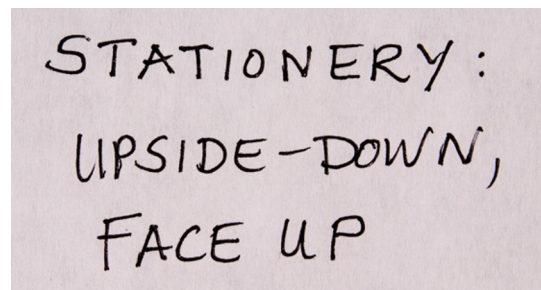


Fig. 30-38

A user-created cognitive affordance to help users know how to insert blank letterhead stationery.

Thus, the user attached his own white adhesive label, shown in [Fig. 30-38](#). As [Norman \(1990, p. 9\)](#) says, “When simple things need pictures, labels, or instructions, the design has failed.”

More interesting and fun examples to come in [Chapter 32](#), on UX design guidelines, but first, we stop and visit the Interaction Cycle in the next chapter.